

Exploring Artificial Intelligence: Concepts, Applications, and Interactions

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Artificial Intelligence (AI)

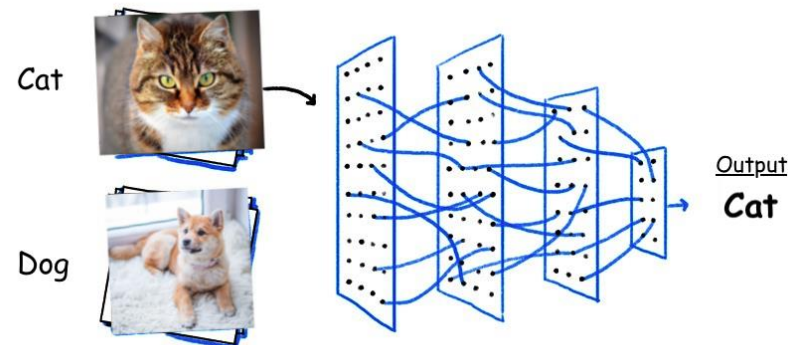
- What is **Artificial Intelligence (AI)**?
 - AI is a branch of Computer Science, Engineering, and Mathematics that aims to create machines and software capable of performing tasks that usually require human intelligence
 - Not one thing. It's hundreds of different things
 - Good idea to talk about a particular function of AI
 - Ex. Deep Neural Networks, Reinforcement Learning, Active Learning, Meta Learning...
 - Technically Machine Learning (ML) is a sub-field of AI, but today encompasses most of AI activities
- Many different **applications** of AI/ML

Reinforcement learning
for Self Driving



vecteezy.com

Deep neural networks for
image recognition/classification



Large language models for
natural language processing



Brief History of Artificial Intelligence

- **Alan Turing:** One of the Founders of Modern CS
 - Universal (Turing) Machine (1937):
 - Turing Test or Imitation Game (1950)
 - To determine if a machine can exhibit intelligent behavior
- **Claude Shannon:** Creator of Information Theory (1940s)
 - Notion of the bit, digital circuits, Boolean algebra
 - Father of the Digital Age
 - No computers/communication systems would be possible without him
- **Dartmouth Conference (1956):**
 - Often called the “Birthplace of AI”
 - A program called **Logic Theorist** was presented
 - Mimic problem solving skills of humans

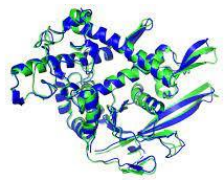


1956 Dartmouth Conference:
The Founding Fathers of AI



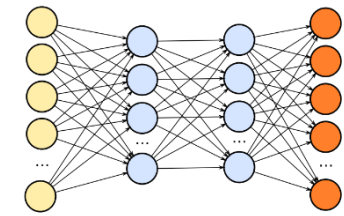
Brief History of Artificial Intelligence

- **1960s-2000s:** Several AI Summers and Winters
 - Great promise followed by disappointment
 - AI never appeared to meet the sky-high expectations
 - Steady progress nonetheless
 - Foundational Advances in ML, expert systems, etc.
 - **Deep Blue (IBM) beats Gary Kasprov...**
- **2010 onwards:** AI starts to go mainstream (**Why?**)
 - Computers become much faster and cheaper
 - Data rich environment → explosion of ML
 - Recommender systems, apps, search engines, all use AI
- **2016-present:** Major Milestones for DNN and RL
 - Alpha-Go beats world champion in the Board Game Go (2016)
 - Alpha-zero uses beats Alpha-Go without human input (2017)
 - AlphaFold “solves” the protein folding problem (2020)
 - Development and release of LLMs like ChatGPT

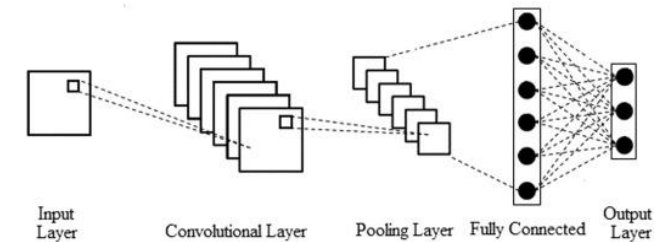


The Pillar of Modern AI: Deep Learning

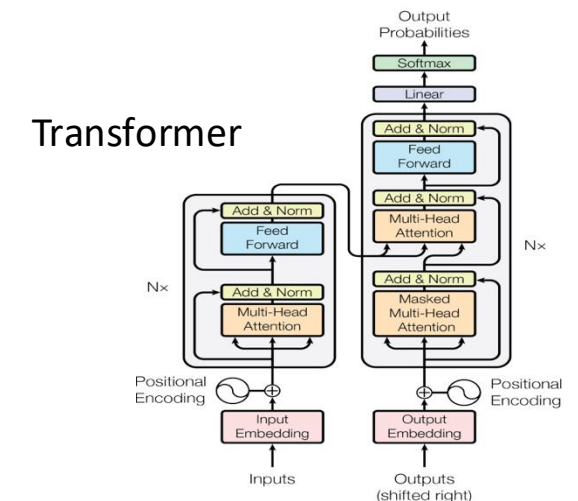
- What is **Deep Learning**?
 - A *subfield of AI (ML)* that uses neural networks with multiple layers “deep” structures to extract high-level features from data
- Typical Components in Deep Learning
 - Large data sets, high-performance computing, and effective training algorithms
 - **Training Phase:** Train the NN on extensive amounts of data to find patterns or features so that it can make accurate predictions on the training data itself
 - **Testing Phase:** Test the NN on new data to see how well it performs on new datasets (generalizability of the NN).
 - If it hasn't done a good job, may need retraining or collect better training data



Neural network



Convolutional neural network



Example: Image Classification for Cats

- We want the **computer** to **recognize a cat!**
- **Training phase:** We need several hundred training images with labels
 - Some images with cats
 - Other images without cats

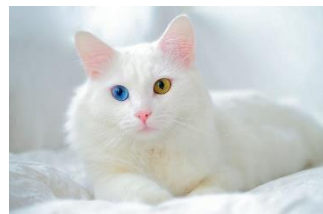


Cat



Not cat

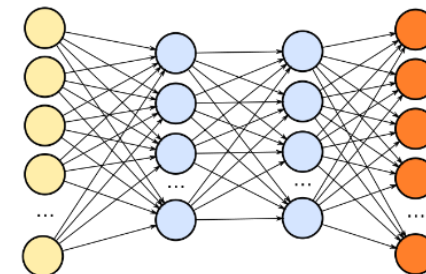
- We feed these images and labels into a neural network



Broken down into a
vector of pixels



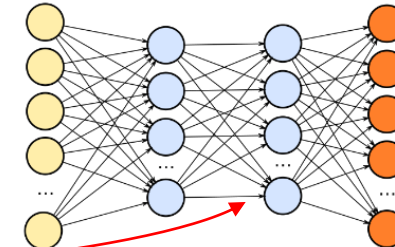
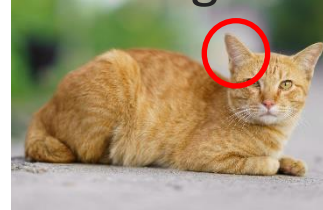
Feed into NN



Example: Image Classification for Cats

- **Deeper Look** at the Training Phase

- NN has lots of **adjustable parts** called **weights**
- For each image, NN makes a guess of whether it is a cat or not, based on the current weights
- If the guess is wrong, then NN will update its weights to correct the guess in the next time
- After training over hundreds of images, NN will develop the ability to recognize specific features
- Training phase typically ends after reaching a **certain error threshold**



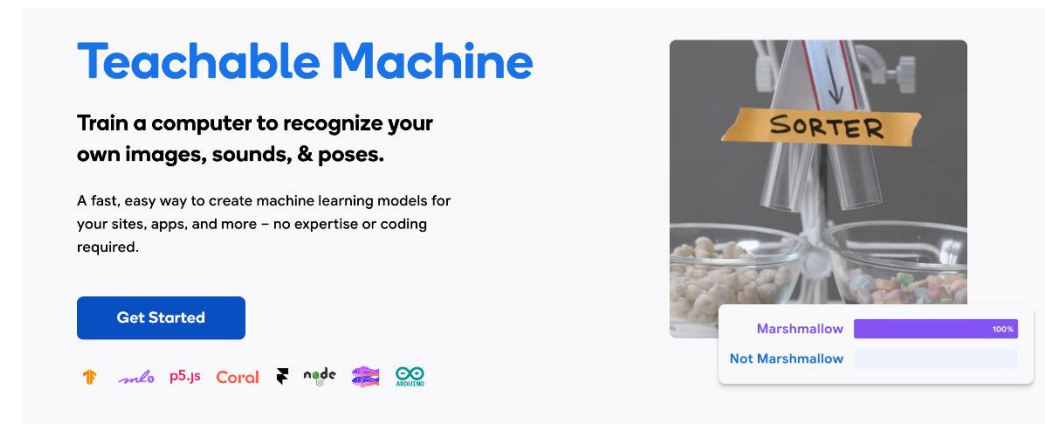
Feature: Pointy ear

- **Testing phase**

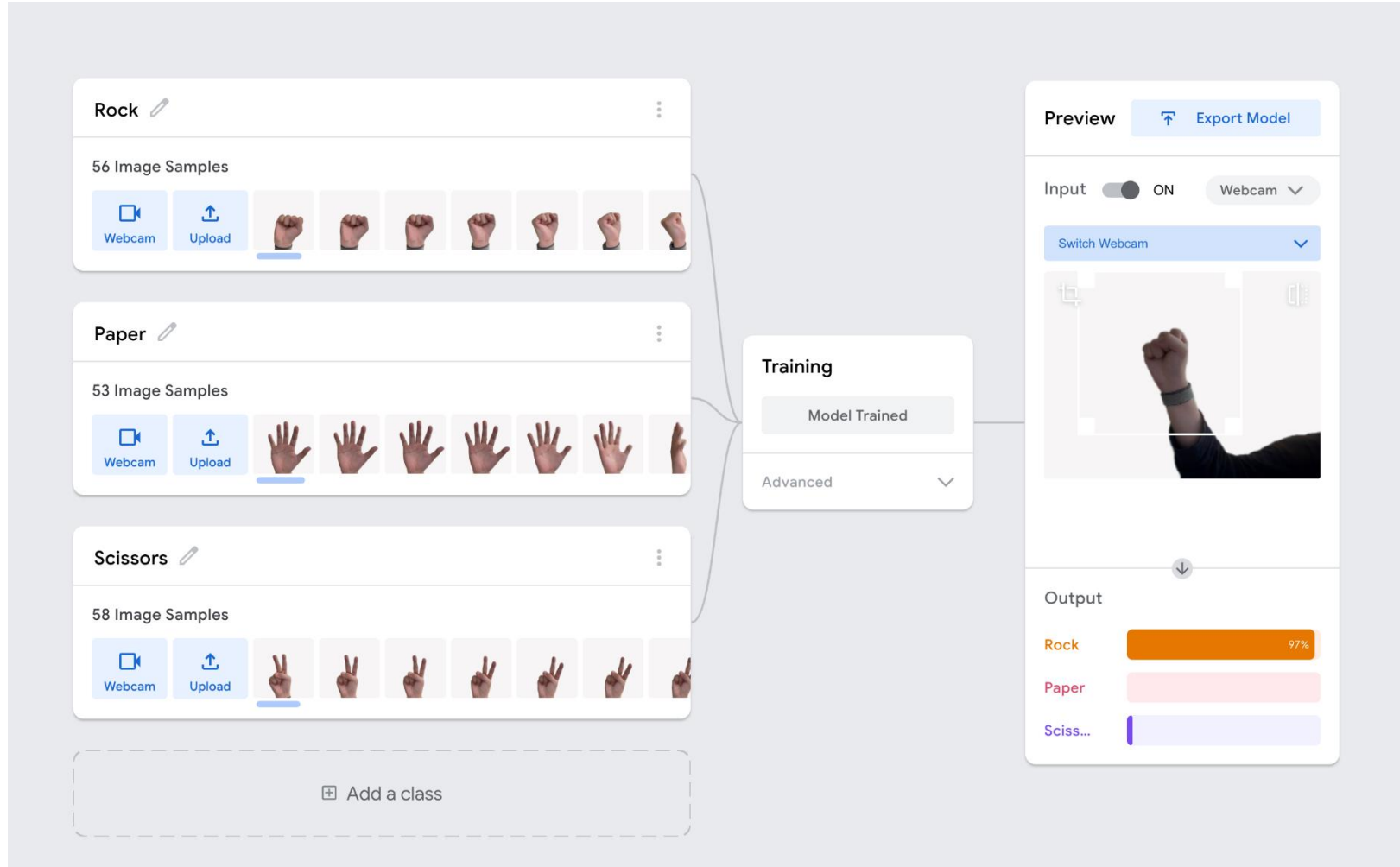
- Ask NN to identify a cat from a new set of images not used in the training phase
- If NN can identify most of them correctly, then our training is successful
 - Otherwise, may need to redo the training with new NN models

Game 1: Image Classification by Teachable Machine

- **What is Teachable Machine?**
 - Lets you teach a computer to recognize **images, sounds, or poses**
 - No coding needed!
 - Works in your web browser
- **Website:** teachablemachine.withgoogle.com
- **Demo:** We'll teach the computer to recognize hand gestures for rock, paper, and scissors



Demo: Gesture Classification



The interface is divided into three main sections: Data Collection, Training, and Preview.

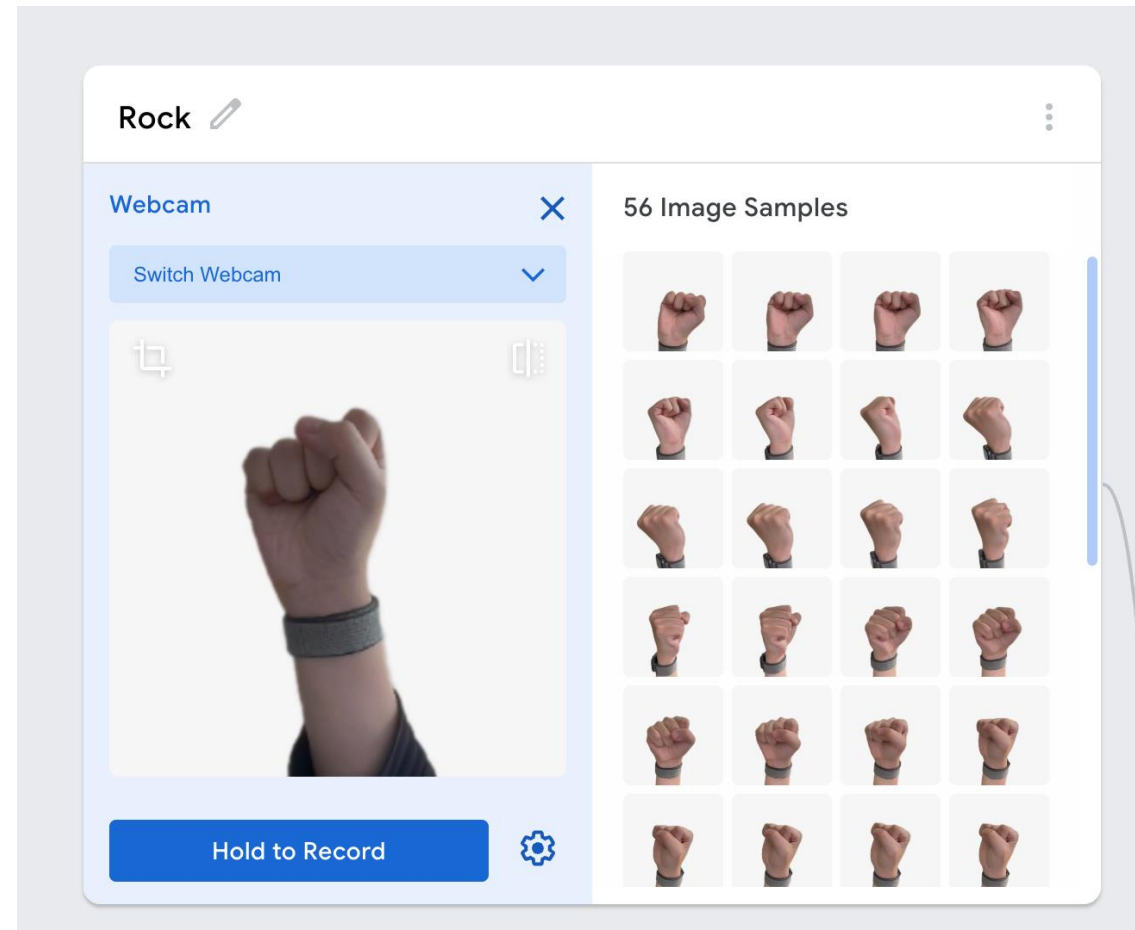
Data Collection: This section allows users to collect image samples for three gesture classes: Rock, Paper, and Scissors. Each class has a 'Webcam' button and an 'Upload' button. Below these buttons are rows of image thumbnails. The 'Rock' class has 56 samples, 'Paper' has 53, and 'Scissors' has 58. At the bottom, there is a dashed box with a plus icon and the text 'Add a class'.

Training: A central panel with a 'Model Trained' button and an 'Advanced' dropdown menu.

Preview: This section shows the model's performance. It includes an 'Input' section with a toggle switch set to 'ON' and a 'Webcam' dropdown. Below this is a 'Switch Webcam' button and a video feed showing a hand making a rock gesture. The 'Output' section displays three horizontal bars representing the model's confidence for each class: Rock (97%), Paper, and Scissors.

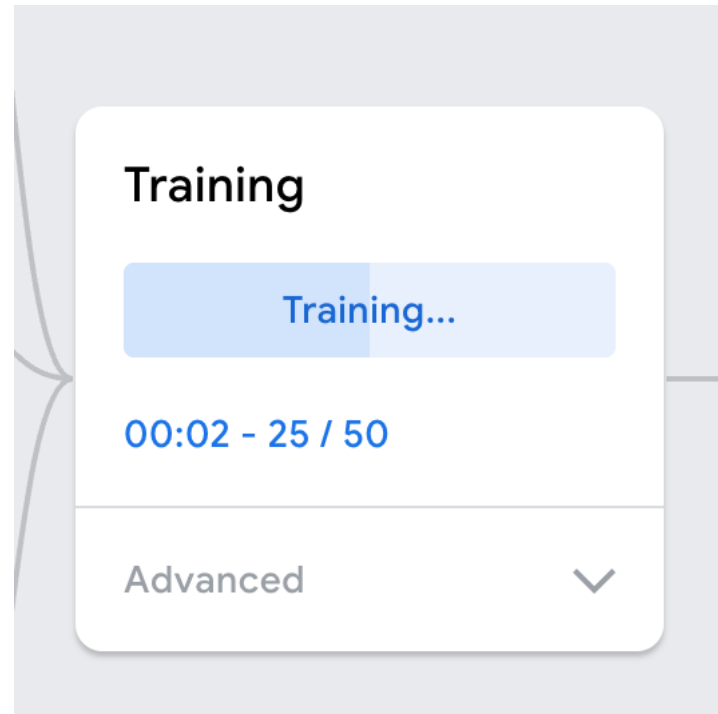
Demo: Prepare the Training Data

- Prepare the data using webcam or collected images



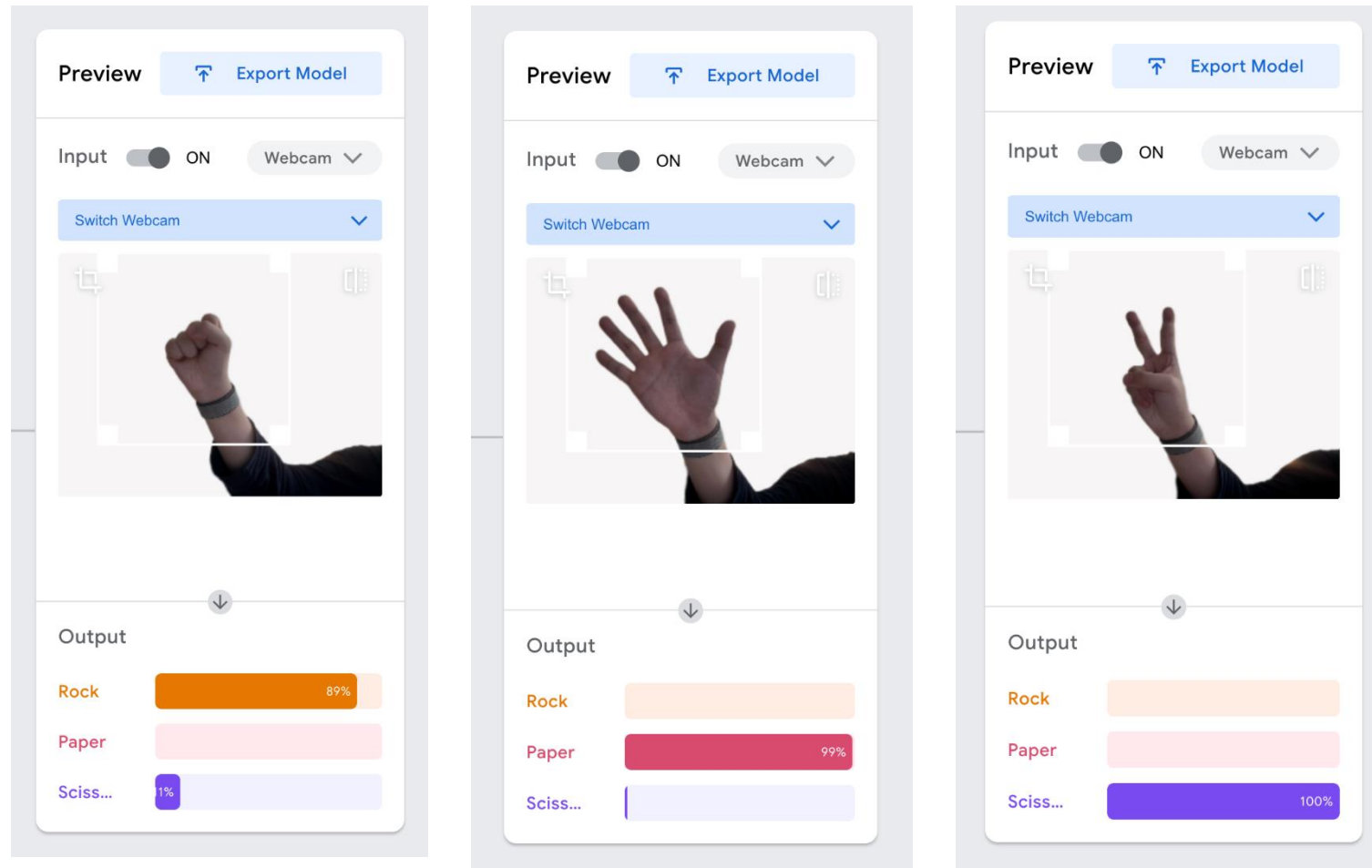
Demo: Train the Model

- Train the model using the prepared dataset



Demo: Inference

- Use the trained model to recognize hand gestures for rock, paper, and scissors



Hand-on Project

- **Dataset: Tiny-Imagenet** <https://tinyurl.com/yrz5zht7>
- 100,000 images of 200 classes (500 images for each class) downsized to 64×64 colored images. (Original Imagenet: at least 224 pixels width/height)
- Each image around 2 KB (dataset total size: ~200 MB)
- You will find 5 folders in the downloaded folder: car, orange, teddy, cat, and dog (with folder names specified). These are called the **training data**
- You will find a test folder, which contains the **testing data**
- Note: complete dataset is [on Huggingface](#)

Dataset: Tiny-Imagenet



Object Localization

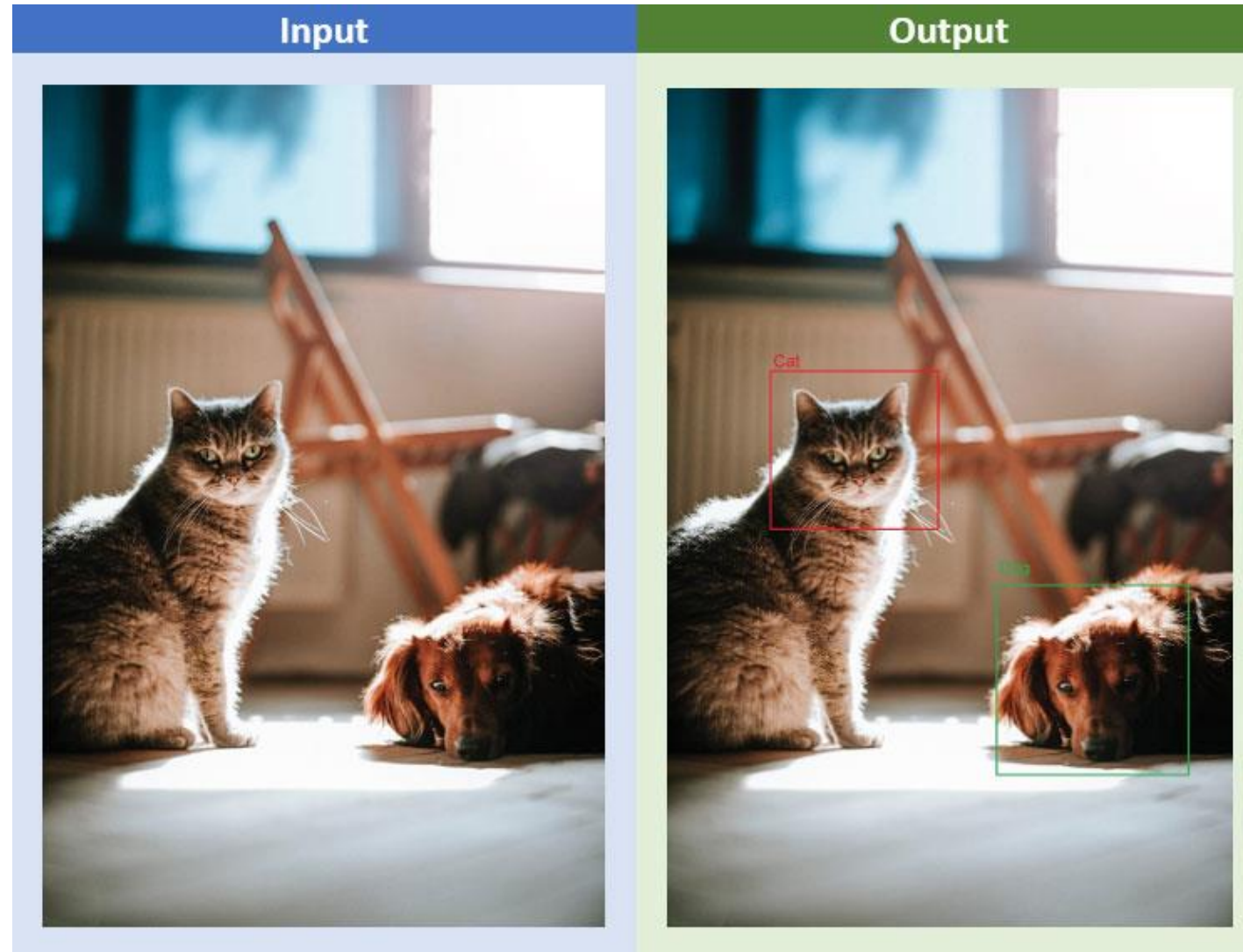


Image Segmentation



(Video) Object Tracking



From YOLOv8

Where can all these be used?

- **Medical/Health care:** analyze medical images (X-rays, MRIs, CT scans) to assist in the early detection and diagnosis of diseases
- **Traffic/Autonomous Vehicle:** identify traffic signs, pedestrians, other vehicles...
- **Agriculture:** monitor crop health, identify diseases, pests...
- **Security:** identify potential threats or suspicious activities

Chatting with AI

- **What is ChatGPT?**
- Developed by OpenAI; GPT-4 released in 2023
- Considered as one of the most advanced AIs right now
- Trained on tons of text from the internet
- Can write stories, answer questions, explain stuff, and more



How do ChatGPT work?

- Generate text by predicting the next word in a sentence
 - Use the context of the conversation - the preceding words and sentences - to make these predictions by using a transformer-like learning architecture
 - Do not possess consciousness or beliefs
 - Instead, LLMs mimic the patterns they learned during training

What Can It Do?

- Answer questions
- Help write essays or poems
- Translate languages
- Write code
- Solve math problems (sometimes)
- Be a chatbot buddy

Turing Test Game

- Two answers for the same question: One by human, the other by ChatGPT
- Which one do you think is written by human?

Question: Describe your perfect day.

Answer 1: Sitting at home doing nothing while knowing I have nothing important to do so I don't feel bad for doing nothing.

Human

Answer 2: A perfect day would be spent enjoying nature, having meaningful conversations with loved ones, achieving a small goal, and ending with a good book or movie to unwind.

ChatGPT

Turing Test Game

- Two answers for the same question: One by human, the other by ChatGPT
- Which one do you think is written by human?

Question: What makes a good friend?

Answer 1: A good friend is someone who is loyal, supportive, honest, and always there to listen and uplift you.

ChatGPT

Answer 2: A good friend is willing to help you as much as you are willing to help your friend, so there is balance.

Human

Turing Test Game

- Two answers for the same question: One by human, the other by ChatGPT
- Which one do you think is written by human?

Question: You have six horses and want to race them to see which is fastest. How many rounds do you need? Just give me a number.

Answer 1: 3.

ChatGPT

Answer 2: 1.

Human

Timeline for ChatGPT

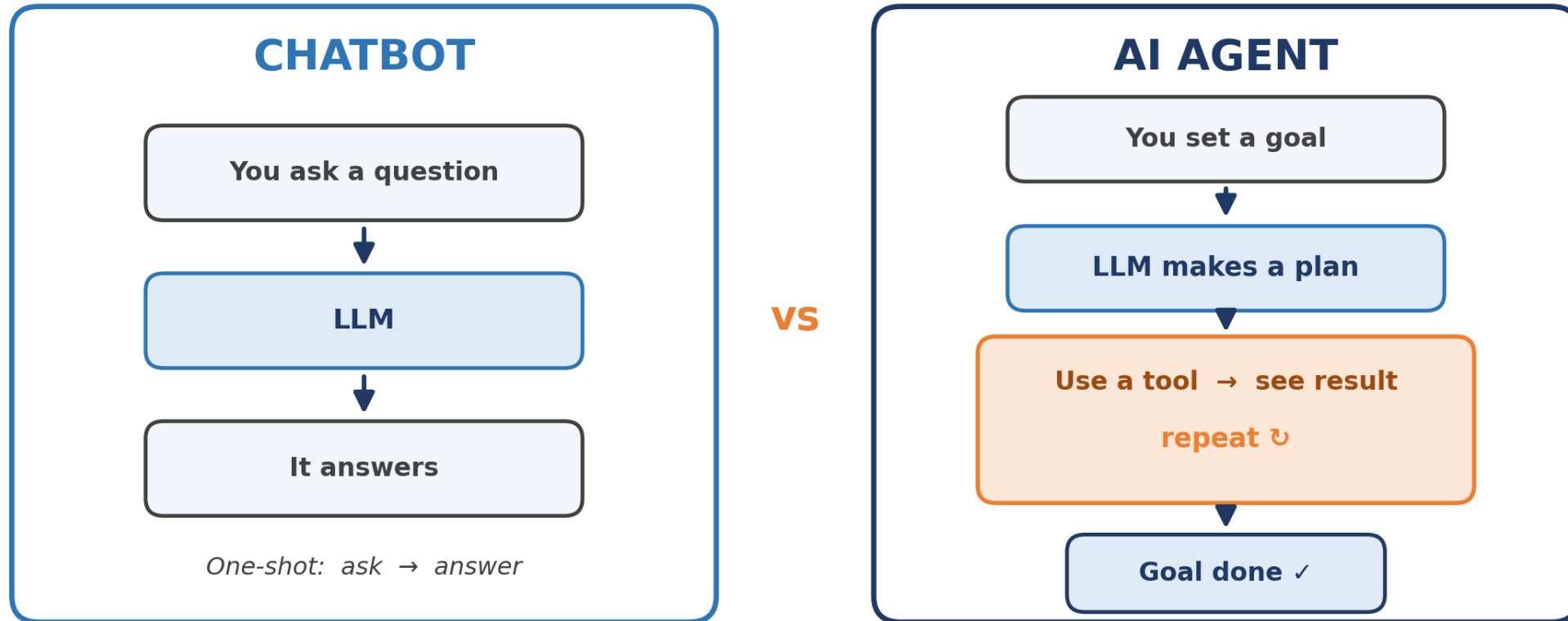
- Jun. 2018, GPT-1 released
 - Large language model with 117 million parameters, the groundwork for GPT series
- Feb. 2019, GPT-2 released
 - 1.5 billion parameters
- Jun. 2020, GPT-3 released
 - 175 billion parameters, very strong in many NLP applications
- Nov. 2022, ChatGPT (GPT-3.5) released, attracted great attention
- Mar. 2023, GPT-4 released
 - Unknown number of parameters, stronger than ChatGPT

Exam	GPT-4	GPT-4 (no vision)	GPT-3.5
Uniform Bar Exam (MBE+MEE+MPT)	298 / 400 (~90th)	298 / 400 (~90th)	213 / 400 (~10th)
LSAT	163 (~88th)	161 (~83rd)	149 (~40th)
SAT Evidence-Based Reading & Writing	710 / 800 (~93rd)	710 / 800 (~93rd)	670 / 800 (~87th)
SAT Math	700 / 800 (~89th)	690 / 800 (~89th)	590 / 800 (~70th)
Graduate Record Examination (GRE) Quantitative	163 / 170 (~80th)	157 / 170 (~62nd)	147 / 170 (~25th)
Graduate Record Examination (GRE) Verbal	169 / 170 (~99th)	165 / 170 (~96th)	154 / 170 (~63rd)
Graduate Record Examination (GRE) Writing	4 / 6 (~54th)	4 / 6 (~54th)	4 / 6 (~54th)
USABO Semifinal Exam 2020	87 / 150 (99th - 100th)	87 / 150 (99th - 100th)	43 / 150 (31st - 33rd)
USNCO Local Section Exam 2022	36 / 60	38 / 60	24 / 60
Medical Knowledge Self-Assessment Program	75 %	75 %	53 %
Codeforces Rating	392 (below 5th)	392 (below 5th)	260 (below 5th)
AP Art History	5 (86th - 100th)	5 (86th - 100th)	5 (86th - 100th)
AP Biology	5 (85th - 100th)	5 (85th - 100th)	4 (62nd - 85th)
AP Calculus BC	4 (43rd - 59th)	4 (43rd - 59th)	1 (0th - 7th)
AP Chemistry	4 (71st - 88th)	4 (71st - 88th)	2 (22nd - 46th)
AP English Language and Composition	2 (14th - 44th)	2 (14th - 44th)	2 (14th - 44th)
AP English Literature and Composition	2 (8th - 22nd)	2 (8th - 22nd)	2 (8th - 22nd)
AP Environmental Science	5 (91st - 100th)	5 (91st - 100th)	5 (91st - 100th)
AP Macroeconomics	5 (84th - 100th)	5 (84th - 100th)	2 (33rd - 48th)
AP Microeconomics	5 (82nd - 100th)	4 (60th - 82nd)	4 (60th - 82nd)
AP Physics 2	4 (66th - 84th)	4 (66th - 84th)	3 (30th - 66th)
AP Psychology	5 (83rd - 100th)	5 (83rd - 100th)	5 (83rd - 100th)
AP Statistics	5 (85th - 100th)	5 (85th - 100th)	3 (40th - 63rd)
AP US Government	5 (88th - 100th)	5 (88th - 100th)	4 (77th - 88th)
AP US History	5 (89th - 100th)	4 (74th - 89th)	4 (74th - 89th)
AP World History	4 (65th - 87th)	4 (65th - 87th)	4 (65th - 87th)
AMC 10 ³	30 / 150 (6th - 12th)	36 / 150 (10th - 19th)	36 / 150 (10th - 19th)
AMC 12 ³	60 / 150 (45th - 66th)	48 / 150 (19th - 40th)	30 / 150 (4th - 8th)
Introductory Sommelier (theory knowledge)	92 %	92 %	80 %
Certified Sommelier (theory knowledge)	86 %	86 %	58 %
Advanced Sommelier (theory knowledge)	77 %	77 %	46 %

GPT-4 Technical Report, OpenAI, Mar. 2023

From Chatbots to AI Agents

ChatGPT answers a question. An AI agent pursues a goal — it plans, uses tools, and acts on its own.



How an AI Agent Works: The Agent Loop

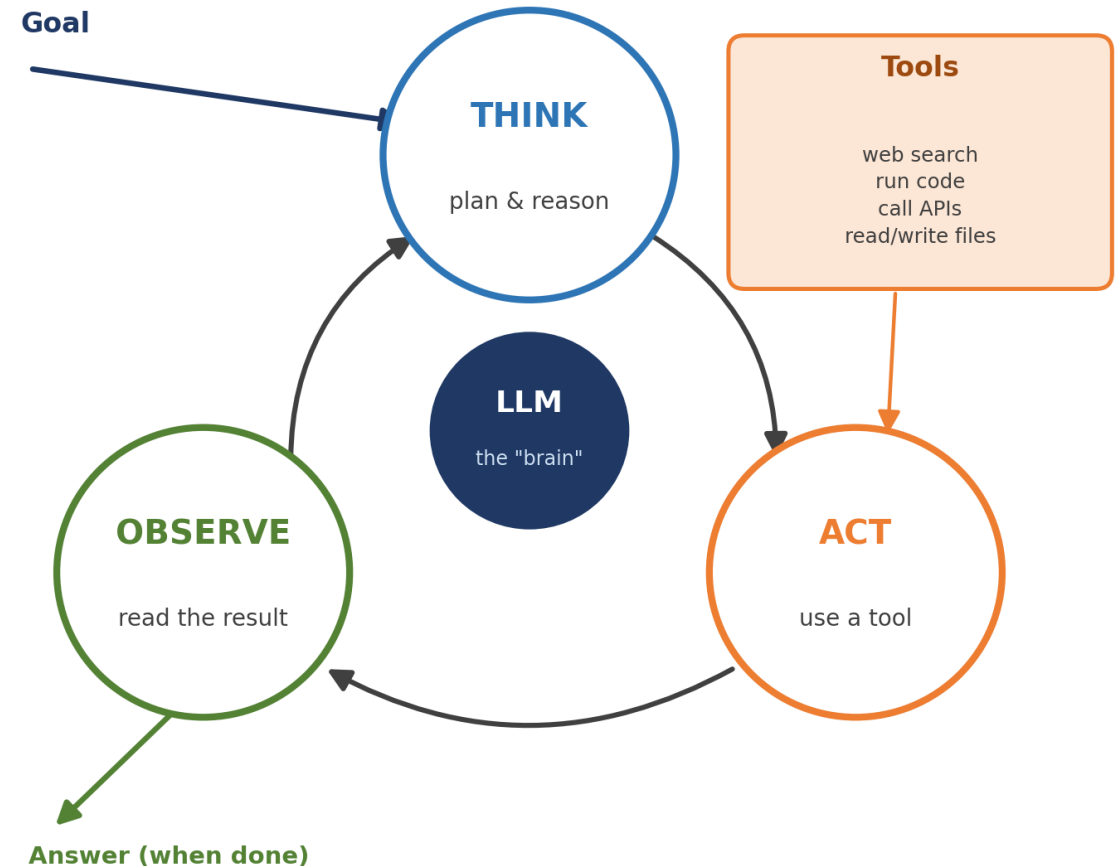
An agent repeats a simple loop:

Think — break the goal into steps

Act — use a tool (search, run code, call an API)

Observe — read the result, then think again

Repeat until the goal is done. The LLM is the brain; the tools are its hands.



AI Agents in the Real World

Agents are already doing real work today:

Coding agents

Write, run & fix code

Cursor, Claude Code, Copilot —
they fix bugs and build whole
features on their own

Research agents

Browse & synthesize

read dozens of web sources,
then write you a report
(e.g. "Deep Research")

Computer-use agents

Click & type like a person

operate real apps & websites
by controlling the screen
(OSWorld benchmark)

Multi-agent teams

Agents that collaborate

a "planner" directs several
worker agents that each
handle a sub-task

The Research Frontier

Powerful, but far from solved — four open problems:



Reliability

Does it finish the job?

agents still slip and get stuck on long, multi-step tasks



Memory

Remember across time

how should an agent recall things across many steps and sessions?



Evaluation

How do we measure it?

what even makes an agent 'good'? surprisingly hard to measure



Safety

Keep it in check

limiting what an autonomous agent is allowed to do on its own

Summary: Key Takeaways

- AI is not one thing — it is hundreds of different techniques
- Machine & Deep Learning: computers learn patterns from data
- The core recipe: Collect data → Train the model → Test on new data
- Computer vision: from recognizing a cat to localization, segmentation & tracking
- ChatGPT predicts the next word — powerful, but it mimics patterns, not real thinking
- The frontier today: AI agents — models that plan, use tools, and act on their own
- AI is already all around us: healthcare, transportation, agriculture, security

Thank You!